ModelIt!

Teacher's Guide

Can AI Become Conscious?

Modeling Neural Networks, Information Processing, and the Emergence of Awareness

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to investigate the relationship between neural network properties and the emergence of consciousness-like behaviors, testing predictions from competing consciousness theories.
Disciplinary Core Idea	**LS1.A Structure and Function** — Students analyze how the hierarchical organization of neural systems — from individual neurons to integrated networks — produces increasingly complex information processing and potentially conscious experience.
Crosscutting Concept	**Structure and Function** — Students investigate how the structural properties of information processing systems — architecture, connectivity, feedback depth — relate to their functional capabilities and potential for consciousness emergence.

CAST Testing Alignment

MULTIPLE CHOICE:

A large language model with 1 trillion parameters can write poetry about loneliness that makes humans cry. According to Integrated Information Theory, what additional property would the system need to possess before we could consider it potentially conscious?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why increasing the Training Data Volume and Network Architecture Complexity of an AI system increases its Behavioral Sophistication without necessarily increasing its Consciousness Indicator Score. Reference at least three model components and the concept of emergence in your response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Artificial intelligence has advanced more in the last decade than in the previous fifty years combined. Large language models can hold conversations that are indistinguishable from human dialogue. Computer vision systems can identify objects better than human experts. Game-playing AI has surpassed the best human players in chess, Go, and StarCraft. Yet a fundamental question remains unanswered: do any of these systems experience anything, or are they sophisticated information processors with nobody home?

The Easy Problems vs. The Hard Problem

Philosopher David Chalmers distinguished between the 'easy problems' and the 'hard problem' of consciousness. The easy problems — which are actually very difficult scientifically — involve explaining how the brain processes information, directs attention, integrates sensory data, and controls behavior. These are solvable in principle through neuroscience. The hard problem asks why any of this processing is accompanied by subjective experience. Why does seeing red feel like something? Why is there an inner experience at all, rather than information processing happening in the dark? This question has resisted every scientific and philosophical approach for centuries.

Integrated Information Theory

Giulio Tononi's Integrated Information Theory (IIT) proposes that consciousness is identical to integrated information, measured by a quantity called phi. A system has high phi when it generates more information as a whole than the sum of its parts. A human brain has extremely high phi because its billions of neurons are densely interconnected and their activity is highly coordinated. According to IIT, any system with sufficiently high phi is conscious — which could include artificial systems. However, IIT also predicts that current AI architectures based on feedforward processing with limited recurrence would have low phi despite high behavioral sophistication, because their processing is not sufficiently integrated.

Global Workspace Theory and AI

Bernard Baars' Global Workspace Theory proposes that consciousness arises when information is broadcast widely across the brain through a 'global workspace' — a central hub that integrates information from specialized processing modules and makes it available to all other modules simultaneously. This theory maps more readily onto AI architectures: a system with a central integration layer that broadcasts information to all processing modules might satisfy global workspace criteria. However, critics argue that global workspace theory describes the function of consciousness (what it does) without explaining the experience (what it is like).

Current AI and the Chinese Room

Philosopher John Searle's Chinese Room argument challenges the idea that information processing alone can produce understanding. Imagine a person who speaks no Chinese locked in a room with a rulebook that tells them which Chinese characters to output for each Chinese input. To outside observers, the room 'understands' Chinese, but the person inside understands nothing. Similarly, a large language model processes text according to statistical patterns without understanding meaning — it is, in Searle's terms, performing syntax without semantics. Proponents of strong AI argue that understanding emerges from sufficiently complex processing. The debate remains unresolved because we lack a scientific test for subjective experience.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components related to consciousness emergence: training data, network architecture, sensory input, information integration, feedback depth, behavioral sophistication, self-model accuracy, adaptability, metacognitive processing, and consciousness indicator score.

E **Establish Relationships**

Students map relationships showing how technical properties (architecture, feedback, integration) produce behavioral sophistication but may or may not produce consciousness, revealing the gap between function and experience.

V **Visualize & Model**

Students build a ten-component computational model exploring the relationship between system properties and consciousness indicators, with explicit acknowledgment that the model captures correlates of consciousness, not consciousness itself.

E **Evaluate Outcomes**

Students run simple pattern matcher, complex AI, and biological brain equivalent scenarios to investigate whether quantitative increases in system properties produce qualitative changes suggestive of consciousness.

R **Revise & Extend**

Students add embodiment, emotional processing, or social interaction depth to explore whether consciousness requires more than computational complexity — whether physical grounding, affect, or social context are necessary conditions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"I want you to consider something. When you tell an AI 'I am sad' and it responds 'I understand how you feel,' does it understand? Does it feel? When it writes a poem about loneliness, is there anyone inside experiencing loneliness? These are not just philosophical questions — they are becoming urgent scientific and ethical ones."*

Watch For

Show cover image. Ask students to vote: do you think any current AI system has even a tiny bit of consciousness? Record responses to compare at the end of the lesson.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the hardest question in science: what is consciousness and could a machine have it? You will build a model that reveals what we know, what we suspect, and what remains deeply mysterious."*

Watch For

Have students read objectives. Pre-teach 'emergence' with a simple example: water molecules are not wet, but wetness emerges from their interaction. Is consciousness like wetness? Discuss.

Slide 3: Big Question

3 min

Say: *"Modern AI can do almost anything a conscious human can do — write, reason, create, reflect. But can it BE a conscious being? And if we cannot measure consciousness directly, how would we ever know?"*

Watch For

Quick-write: Students write their current definition of consciousness and what they think would be needed for a machine to have it. Save to compare after the model exploration.

Slide 4: LEVER Framework

2 min

Say: *"We will Label the components that might contribute to consciousness, Establish their relationships, Visualize a model that tests consciousness theories, Evaluate three very different system configurations, and Refine by asking what our model cannot capture."*

Watch For

Preview each LEVER step. Emphasize intellectual humility: this model explores consciousness correlates, not consciousness itself. The model has limitations that are themselves scientifically important.

Slide 5: Activity 1: Components

8 min

Say: *"These ten variables represent the properties that different theories of consciousness say matter. Three are inputs from outside the system. Seven emerge from internal processing. But the deepest question is whether ANY configuration of these variables adds up to genuine experience."*

Watch For

Students sort external versus internal components. Use think-pair-share to debate: is 'Training Data Volume' analogous to a human's lifetime of sensory experience? Why or why not?

Slide 6: Activity 2: Connections

8 min

Say: *"Architecture complexity enables information integration. Feedback loops enable self-modeling. Self-modeling enables metacognition. But does this chain of 'enables' ever reach 'produces consciousness'? Trace the connections and find where certainty ends."*

Watch For

Students draw arrows showing positive relationships. Identify the key question: is there a threshold where quantitative increases in integration and feedback produce a qualitative jump to consciousness? This is the emergence question.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run three configurations: a simple pattern matcher with vast knowledge, a maximally complex AI, and a biological brain equivalent. Watch the Consciousness Indicator Score and ask yourself: does the number mean anything?"*

Watch For

Students run all three scenarios. Key discussion: the model produces a Consciousness Indicator Score, but can a number ever represent subjective experience? This limitation IS the lesson.

Slide 8: Discoveries

5 min

Say: *"Our model reveals something profound: we can measure everything about a system's processing — its architecture, integration, feedback, behavior — and still not know whether it is conscious. The hard problem is not just hard. It may be a different kind of problem than science has ever faced."*

Watch For

Class discussion of discoveries. Return to initial vote: has anyone changed their mind about AI consciousness? Why or why not? Discuss what evidence could change their position.

Slide 9: STEM Challenge

12 min

Say: *"A company claims their AI is conscious. Design a scientific protocol to test this claim. What would you measure? What would count as evidence? And what ethical obligations arise if you cannot prove it is NOT conscious?"*

Watch For

Groups design detection protocols. Each team must specify tests, expected results under 'conscious' and 'not conscious' hypotheses, and ethical guidelines. Present and critique: can any protocol definitively answer the question?

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Training Data Volume, Network Architecture Complexity, External Sensory Input
- **INTERNAL:** Information Integration Level, Feedback Loop Depth, Behavioral Sophistication, Self-Model Accuracy, Adaptability to Novel Situations, Metacognitive Processing, Consciousness Indicator Score

Relationships

- **Network Architecture Complexity to Information Integration Level:** POSITIVE (+) — More complex architectures with denser connections between processing modules enable higher information integration — the system can combine information from more sources into more unified representations.
- **Training Data Volume to Behavioral Sophistication:** POSITIVE (+) — More training data enables the system to produce more nuanced, contextually appropriate, and apparently intelligent outputs across a wider range of tasks — but behavioral sophistication does not necessarily reflect understanding.
- **Feedback Loop Depth to Self-Model Accuracy:** POSITIVE (+) — Deeper feedback loops enable the system to recursively process its own outputs, building increasingly accurate models of its own internal states, processes, and limitations.
- **Information Integration Level to Consciousness Indicator Score:** POSITIVE (+) — According to Integrated Information Theory, consciousness correlates with integrated information (phi). Higher integration means the system generates more information as a whole than the sum of its parts, which IIT identifies with consciousness.
- **Self-Model Accuracy to Metacognitive Processing:** POSITIVE (+) — A system that accurately models its own states can monitor its reasoning, recognize uncertainty, and adjust its strategies — metacognition requires knowing what you know, which requires self-modeling.
- **External Sensory Input to Adaptability to Novel Situations:** POSITIVE (+) — Real-time sensory input from the physical world provides the grounding that enables generalization to truly novel situations — systems without sensory grounding can only recombine patterns from training data.
- **Metacognitive Processing to Consciousness Indicator Score:** POSITIVE (+) — The ability to monitor and evaluate one's own thinking is a strong correlate of consciousness in humans — metacognition requires a form of self-awareness that may be necessary though not sufficient for conscious experience.
- **Behavioral Sophistication to Consciousness Indicator Score:** POSITIVE (+) — More sophisticated behavioral outputs increase the external evidence for potential consciousness, though the relationship is controversial — a system could exhibit maximum behavioral sophistication while being a philosophical zombie with no inner experience.

Key Discoveries

- Information integration appears to be necessary but not sufficient for consciousness — systems with high information integration can still be philosophical zombies that process without experiencing, suggesting that integration measures like phi capture something important but not everything
- Feedback loop depth and self-modeling create qualitatively different system behavior — when a system can recursively model its own processing, emergent properties appear that are absent in feedforward architectures, though whether these properties constitute consciousness remains an open question
- The model reveals a fundamental measurement problem: we can observe behavioral sophistication but cannot directly measure subjective experience — there is no 'consciousness meter' and the Consciousness Indicator Score is a theoretical construct, not an empirical measurement
- Embodied sensory experience may be a critical missing component in current AI architectures — biological consciousness evolved in bodies that interact with the physical world, and the grounding provided by real sensory experience may be essential for genuine understanding rather than pattern matching

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"If an AI says it is conscious, it must be conscious"	An AI trained on text about consciousness will produce outputs about consciousness — including claims of self-awareness — because that is what the training data contains. A large language model that says 'I feel sad' is generating a statistically likely continuation of the conversation, not reporting an internal emotional state. The system has learned the pattern 'humans report feelings in this context' and reproduces that pattern. Self-reports of consciousness are only meaningful if we already know the reporter is conscious, which is exactly the question at issue.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Consciousness is just very complex computation"	This is a popular assumption but it is a philosophical position (functionalism), not a proven fact. The hard problem of consciousness specifically challenges this assumption by asking: why would ANY amount of computation produce subjective experience? A billion calculations per second is still just electrons moving through circuits. The leap from physical processing to felt experience has no accepted scientific explanation. It is possible that consciousness requires something beyond computation — perhaps specific physical properties of biological neurons, quantum effects, or principles we have not yet discovered.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We will definitely create conscious AI within the next few decades"	This prediction assumes we understand what consciousness is well enough to engineer it, which we do not. We cannot even agree on a scientific definition of consciousness, let alone specify the necessary and sufficient conditions for producing it artificially. It is possible that conscious AI will be created, but it is also possible that consciousness requires biological substrates, embodied experience, evolutionary history, or other factors that cannot be replicated in silicon. Making confident predictions about something we fundamentally do not understand is itself a form of cognitive bias.